KOLEJ MATRIKULASI PULAU PINANG

PRE PSPM
DP014
SET 1

2023/2024

Instruction:

This question paper consists of 7 subjective questions.

Attempt all questions .

QUESTION	MARKS
1	/ 8
2	/ 12
3	/ 9
4	/ 12
5	/ 9
6	/ 12
7	/ 18
TOTAL	/ 80

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $p = mv$
6. $J = F\Delta t$
7. $J = \Delta p = mv - mu$
8. $f = \mu N$
9. $W = Fs\cos\theta$
10. $K = \frac{1}{2}mv^2$
11. $U = mgh$
12. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
13. $s = r\theta$
14. $v = r\omega$

16. $a_c = \frac{v^2}{r} = r\omega^2$
17. $\omega = \omega_o + \alpha t$
18. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$
19. $\omega^2 = \omega_o^2 + 2\alpha\theta$
20. $\theta = \frac{1}{2}(\omega_o + \omega)t$
21. $\tau = rF\sin\theta$
22. $\frac{dQ}{dt} = -kA\left(\frac{dT}{dx}\right)$
23. $v_{rms} = \sqrt{\frac{3RT}{M}}$
24. $v_{rms} = \sqrt{\frac{3kT}{m}}$
25. $Q = \Delta U + W$

15. $a_t = r\alpha$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space *Ketelapan ruang bebas*	μ_0	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space *Ketelusan ruang bebas*	ε_0	$= 8.85 \times 10^{-12}$ F m^{-1}
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}$ C
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}$ J s
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}$ kg $= 5.49 \times 10^{-4}$ u
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}$ kg $= 1.008665$ u
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}$ kg $= 1.007277$ u
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}$ kg $= 2.014102$ u
Molar gas constant *Pemalar gas molar*	R	$= 8.31$ J K^{-1} mol^{-1}
Rydberg constant *Pemalar Rydberg*	R_H	$= 1.097 \times 10^{7}$ m^{-1}
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Gravitational constant *Pemalar graviti*	G	$= 6.67 \times 10^{-11}$ N m^{2} kg^{-2}
Free-fall acceleration *Pecutan jatuh beba*	g	$= 9.81$ m s^{-2}

1. (a) (i) The maximum speed of a Boeing 747 is 988 km h^{-1}.

What is its maximum speed in m s^{-1}?

(ii) A human hair has a thickness of 60 µm. Convert this value to meters

[2 *marks*]

(b) Three forces A, B and C act on an object as shown in **FIGURE 1** below.

Find the resultant force of these forces.

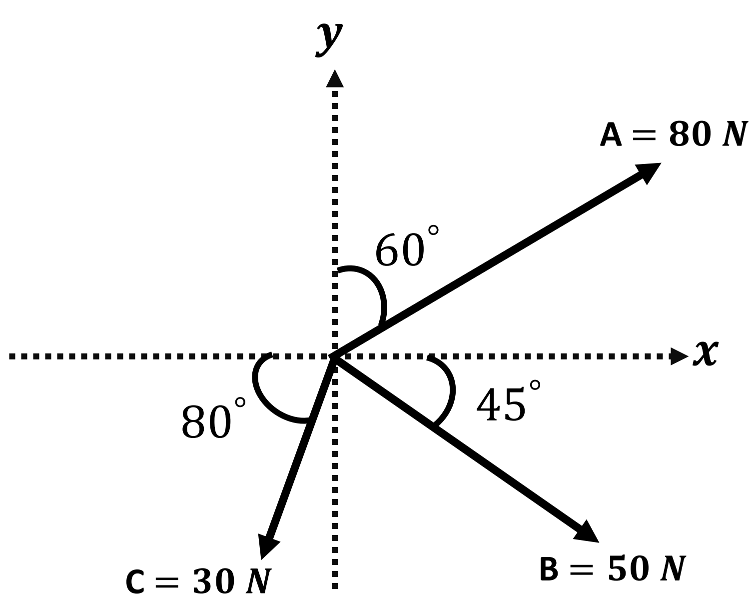

FIGURE 1

[6 *marks*]

2(a)

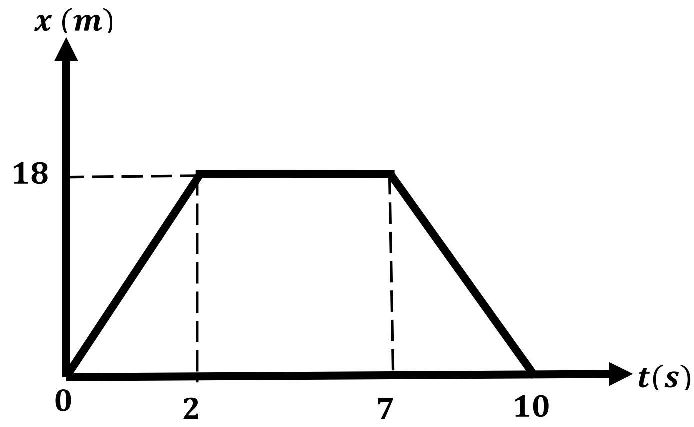

FIGURE 2

FIGURE 2 shows a graph of displacement, *x* against time, *t* for a car travelling along a straight road.

(i) Calculate the velocity of the car for the time 0-2 s, 2-7 s and 7-10 s.

(ii) Sketch a velocity against time graph for the whole journey of the car.

[7 *marks*]

(b) An object moves in a straight line with uniform acceleration of 2 m s^{-2} over a distance of 30 m in 3 s. Calculate the initial velocity of the object.

[2 *marks*]

(c) A ball is thrown straight upward with an initial velocity of 15 m s^{-1}.
Determine the magnitude and the direction of the ball's displacement after 2 s it is thrown.

[3 *marks*]

3.

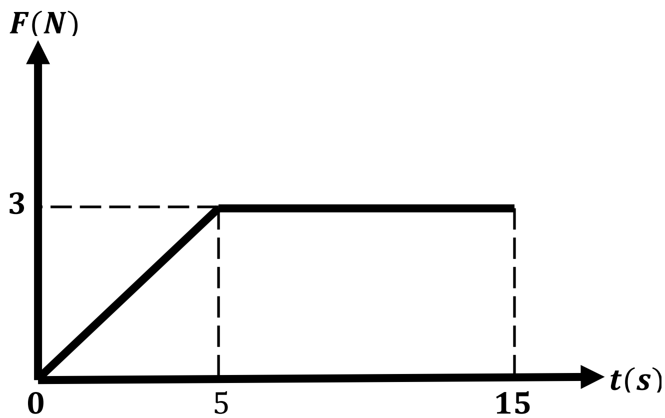

FIGURE 3

a) **FIGURE 3** shows a graph of force F acting on a 5.0 kg ball that varies with time, t. The ball is initially at rest. Determine

i) the impulse on the ball,

ii) the speed of the ball after 15 s.

[2 *marks*]

b) A 2000 kg truck moving to the right at speed of 80 m s^{-1} strikes a stationary car with mass of 860 kg. After the collision, the truck rebounds with a speed of 5 m s^{-1} at an angle of 30^{O} above +ve *x* axis. Determine

i) the magnitude of x component final velocity of the car.

ii) the magnitude of y component final velocity of the car.

iii) the magnitude of final velocity of the car.

[7 *marks*]

4a.

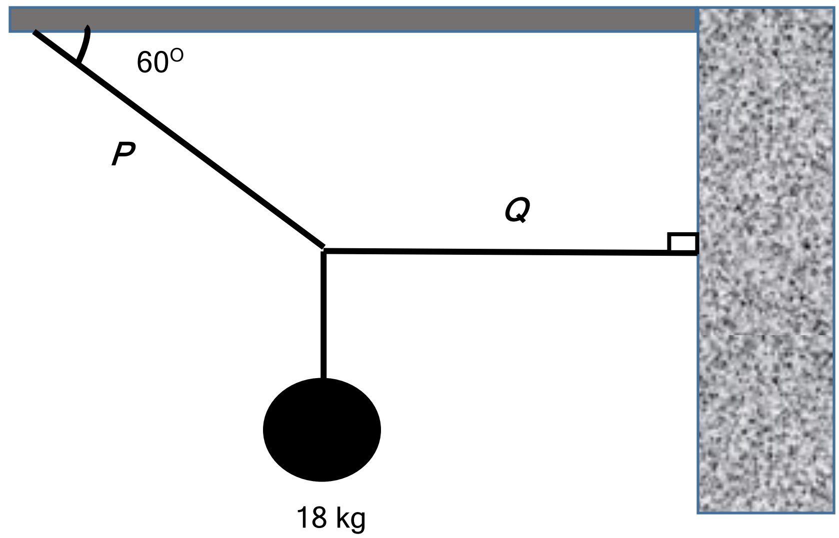

FIGURE 4

FIGURE 4 shows a 18 kg load held in equilibrium by ropes, ***P*** and ***Q*** fastened to the ceiling and the wall respectively. Calculate the tension ***P*** and ***Q***. [4 *marks*] b.

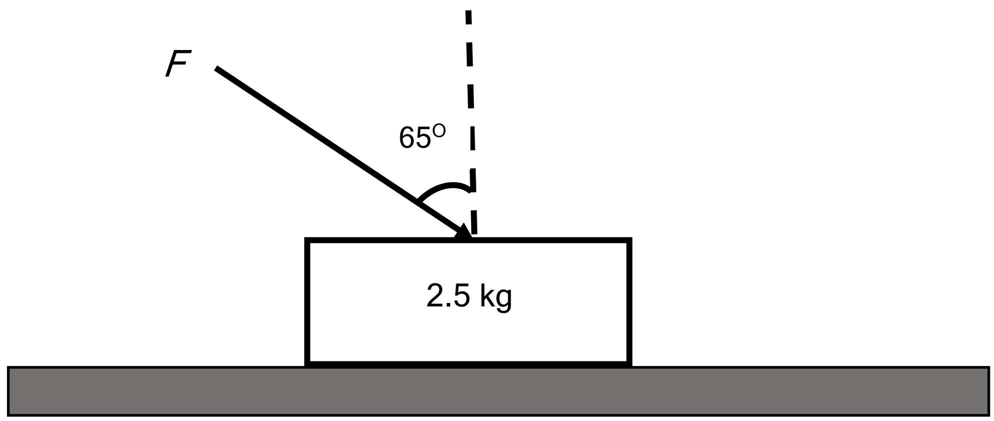

FIGURE 5

FIGURE 5 shows a 2.5 kg block is being pushed along a rough horizontal surface by a force F = 43 N at an angle 65° from the normal. The block moves at a constant acceleration 0.6 ms^{-2}.

i. Sketch a free body diagram for the block.

ii. Calculate the frictional force.

iii. Determine the coefficient of friction between the block and the rough surface.

[8 *marks*]

5 (a) An object completes 95 cycles in 155 s. Calculate the

(i) period.

(ii) angular displacement in 5 minutes. [3 *marks*]

(b) The coefficient of friction between a rider and the merry go round is 0.45 and the person is measured to be traveling at 20.0 revolutions per minute.

(i) Convert the angular velocity in SI unit.

(ii) Calculate the radius of the person be standing. [3 *marks*]

(c)

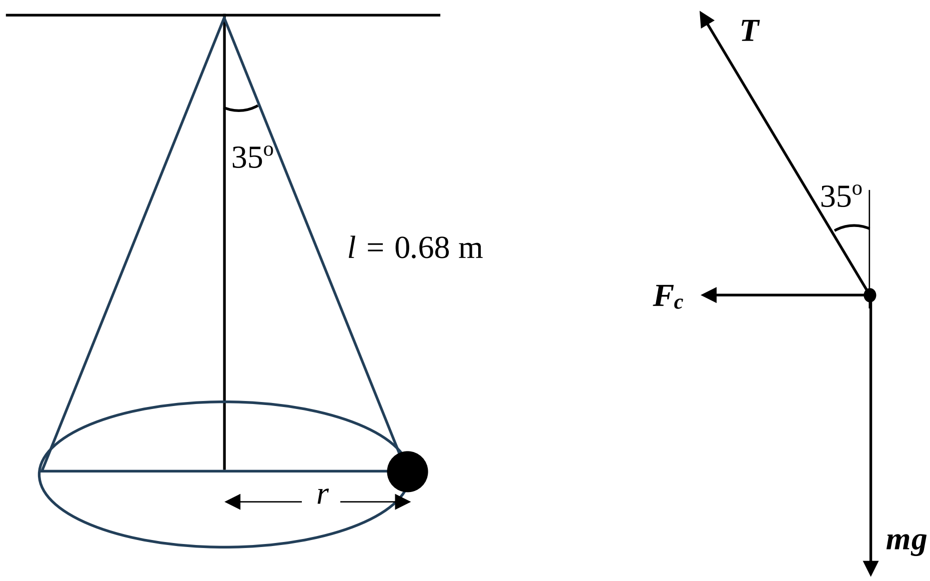

FIGURE 6

FIGURE 6 shows a 2.5 kg mass is rotated into a conical pendulum where the length of string connected to the mass is 0.68 m with the angle between the string and vertical is 35° and its free body diagram.

Calculate the

(i) radius of the circle.

(ii) the speed of the mass. [3 *marks*]

6(a). A blade of a ceiling fan has a radius of 0.400 m is rotating about a fixed axis with an initial angular velocity of 0.150 $revs^{-1}$.The angular acceleration of the blade is 0.750 $revs^{-2}$. After 4.00 seconds, determine;

i) the angular velocity,

ii) the number of revolutions for the blade turns,

iii) the tangential speed of a point on the edge of the blade. [6 *marks*]

(b). If the blade that rotates at the same initial angular velocity decelerating at the same rate, determine the time taken by the blade to completely stop rotating. [2 *marks*]

(c)

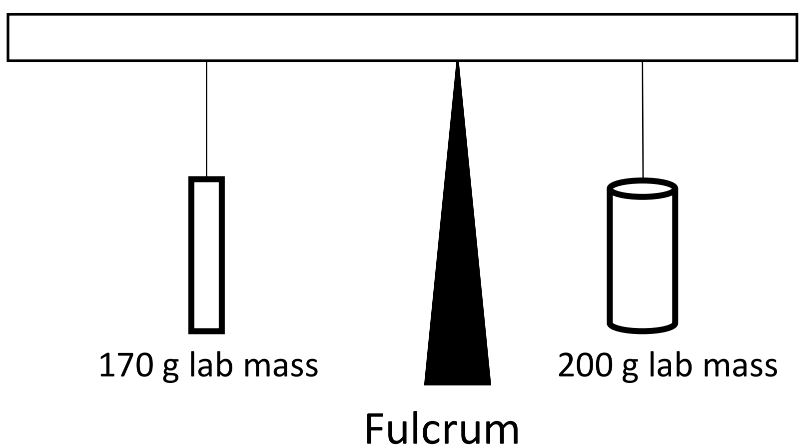

FIGURE 7

FIGURE 7 shows a 200 g mass is hung from a weightless meter stick, 20 cm from the fulcrum. A 170 g mass is used to balance the system. How far will it have to be located from the fulcrum to keep the system in balance?

[4 *marks*]

7(a) A perfectly insulated aluminium rod has length of 40 cm and cross sectional area of $4.0\ cm^2$. At a steady state, the temperature at both ends are $50^{\circ}C$ and $150^{\circ}C$ respectively. Given the thermal conductivity of aluminium is $210\ Wm^{-1}K^{-1}$.

Calculate the

(i) temperature gradient along the rod.

(ii) rate of heat flow.

[4 *marks*]

(b) The rms speed of a gas at a temperature of $25^{\circ}C$ is $600\ ms^{-1}$. Calculate the

(i) mass of an atom of the gas.

(ii) temperatue of the gas if the rms speed increased to $750\ ms^{-1}$.

[4 *marks*]

(c) A gas which undergoes an adiabatic process does 5.0 kJ of work.

(i) Calculate the change in its internal energy.

(ii) In another situation, if the change of internal energy of the gas is 3.0 kJ, calculate the work done.

[5 marks]

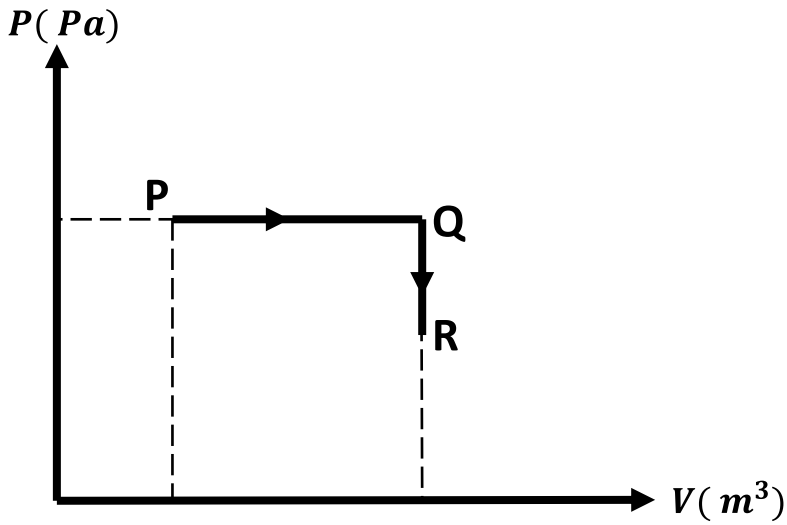

FIGURE 8

(d) An ideal gas undergoes two processes PQ and QR as shown in **FIGURE 8**.

10 kJ of heat is supplied to the process PQ and 6 kJ of work is done when the gas expands from P to Q. A 2 kJ of heat is dissipated from the gas in the QR process.

Calculate the

(i) change in the internal energy for PQ.

(ii) change in the internal energy for QR.

[5 *marks*]

SOALAN TAMAT